Power

File: power.kicad_sch

Level 2 drive

File: level2.kicad_sch

Head

File: head.kicad_sch

Piper-Watch wiring. Power: the 19 V feed, split block, panel fuse, switch and buck.

Level 2 drive: the perfboard (TMC2209), base ESP32-S3, Hall sensor and motor. Head: the XIAO and the three displays.

Nets that cross sheets use global labels: +19V, +19V_FUSED, +19V_SW, +5V, GND. Everything else is local to its sheet.

USB (data + logic power) runs to the Jetson from: base ESP32-S3, head XIAO ESP32-S3, C920 camera board.

Wiring draft generated from hardware/wiring.md

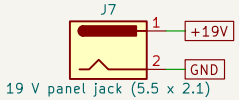
Piper-Watch

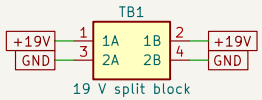
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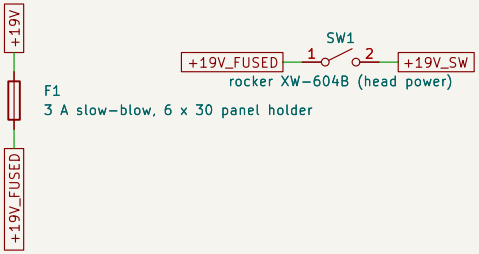
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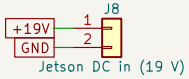
Title: Piper-Watch wiring

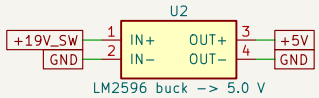
Size: A4	Date:	**Rev:**
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19 V supply -> panel jack -> split block TB1 (on Level 2's floor). TB1's − side is the COMMON GROUND.

The Jetson is fed straight from TB1: always on, never switched or fused here (the supply limits it).

Head side: TB1 + -> panel fuse F1 (6 x 30 mm, 3 A slow-blow) -> rocker switch SW1 -> perfboard J1 (+19V_SW).

Fuse holder: TB1 to its END contact, its SIDE contact to the switch (an unscrewed cap's exposed end is dead).

U2 (LM2596 buck, on its own floor bosses) is fed from perfboard J6; set it to 5.0 V before connecting the Hall sensor.

Wiring draft generated from hardware/wiring.md

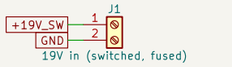
Piper-Watch

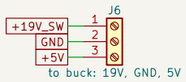
Sheet: /Power/

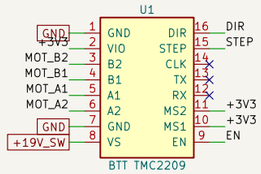
File: power.kicad_sch

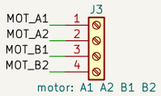
Title: Power: 19 V in, split block, panel fuse, switch, 5 V buck

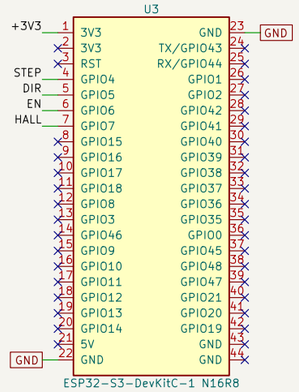
Size: A4	Date:	**Rev:**
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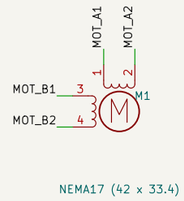




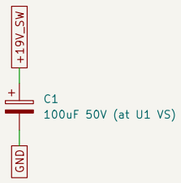


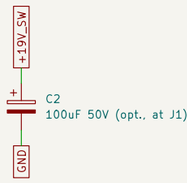


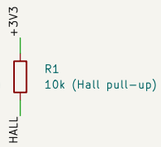




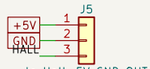


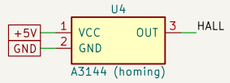












PERFBOARD (70 x 50): screw terminals J1 (19 V in), J3 (motor), J6 (buck); pin headers J4 (ESP32), J5 (Hall) with Dupont housings.

Heavy wire (20–22 AWG): 19 V, GND, motor. C1 sits right at U1's VS and the GND next to it; C2 (optional) at J1.

U1 MS1 = MS2 = 3V3: 1/16 microstepping. CLK, TX, RX and DIAG unconnected. Set Vref ~0.7 V before connecting the motor.

R2 (optional) holds EN high (motor off) while the ESP32 boots.

Motor M1: 6-pin socket, pins 1+3 = coil A (red, blue) -> A1, A2; pins 4+6 = coil B (green, black) -> B1, B2; pins 2, 5 (centre taps) unused.

U3 (base ESP32-S3) is powered by its USB cable from the Jetson — never also from the buck's 5 V.

Hall U4 (A3144) on the motor bracket's pad: VCC from the 5 V buck, OUT pulled up to 3V3 by R1 (never to 5 V).

Wiring draft generated from hardware/wiring.md

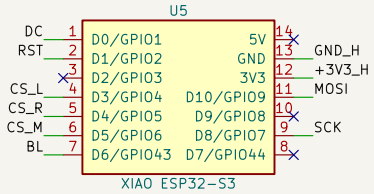
Piper-Watch

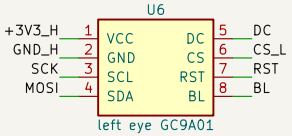
Sheet: /Level 2 drive/

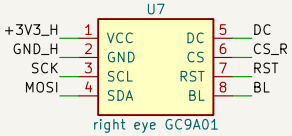
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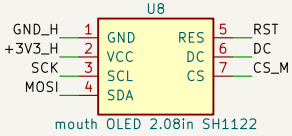
Title: Level 2: perfboard (TMC2209), base ESP32-S3, Hall sensor, motor

Size: A3	Date:	**Rev:**
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The XIAO and the C920 camera each have their own USB cable to the Jetson, through the turntable's hollow.

No Wi-Fi/BT: the XIAO's antenna stays unplugged. Displays run from the XIAO's 3V3 (~150–250 mA total).

One SPI bus (SCL, SDA, DC, RST shared); a CS per display. 26–28 AWG silicone wire soldered straight to the pads.

The eyes mount upside down: the firmware rotates their image 180 deg. D2 (GPIO3, strapping), D7, D9 free.

Wiring draft generated from hardware/wiring.md

Piper-Watch

Sheet: /Head/

File: head.kicad_sch

Title: Head: XIAO ESP32-S3, two GC9A01 eyes, SH1122 mouth

Size: A4	Date:	**Rev:**
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